

## Paper II Audit -- P4: Integer maximization = $\text{floor}((2n+1)^2/24)$

|             |                                                          |
|-------------|----------------------------------------------------------|
| Test        | P4 -- Integer maximization = $\text{floor}((2n+1)^2/24)$ |
| Gate        | F (Prop 7.1 / L11)                                       |
| Evidence    | EXACT integer/Fraction arithmetic, $n=1..5000$           |
| Environment | Python 3.14.4 / Windows-11-10.0.26200-SP0                |

### 1. Claim under test (Prop 7.1)

For  $p+q=n$ ,  $\text{Phi\_tau}(S_{\{p,q\}}) \leq \text{floor}((2n+1)^2/24)$ , with equality in the saturated branch at an integer  $p$  nearest  $(2n+1)/6$ . The universal engine is the complete-the-square identity plus the residue behaviour of odd squares mod 24; both are checked exactly.

### 2. Kill-switch (declared before running)

**H4a** integer max == floor; **H4b** complete-the-square identity exact; **H4c** attainment at nearest integer, saturated branch; **H4d** odd-square mod-24 classification (==1, or ==9 if divisible by 3). Falsify on any mismatch.

### 3. Result

Over  $n=1..5000$  (exact): max mismatches **0**; identity fails **0**; attainment fails **0**; residue fails **0**.

**Verdict: VERIFIED: integer maximum equals  $\text{floor}((2n+1)^2/24)$  for all  $n$  in  $1..5000$  (exact); complete-the-square identity, saturated-branch attainment, and the odd-square mod-24 residue classification all hold.**

### 4. Sample values

| n  | max Phi_tau | $\text{floor}((2n+1)^2/24)$ | equal? | argmax p |
|----|-------------|-----------------------------|--------|----------|
| 1  | 0           | 0                           | yes    | 0        |
| 2  | 1           | 1                           | yes    | 1        |
| 3  | 2           | 2                           | yes    | 1        |
| 4  | 3           | 3                           | yes    | 1        |
| 5  | 5           | 5                           | yes    | 2        |
| 6  | 7           | 7                           | yes    | 2        |
| 7  | 9           | 9                           | yes    | 2        |
| 8  | 12          | 12                          | yes    | 3        |
| 9  | 15          | 15                          | yes    | 3        |
| 10 | 18          | 18                          | yes    | 3        |
| 11 | 22          | 22                          | yes    | 4        |
| 12 | 26          | 26                          | yes    | 4        |
| 13 | 30          | 30                          | yes    | 4        |
| 14 | 35          | 35                          | yes    | 5        |
| 15 | 40          | 40                          | yes    | 5        |
| 16 | 45          | 45                          | yes    | 5        |
| 17 | 51          | 51                          | yes    | 6        |
| 18 | 57          | 57                          | yes    | 6        |
| 19 | 63          | 63                          | yes    | 6        |
| 20 | 70          | 70                          | yes    | 7        |

| n  | max Phi_tau | floor((2n+1)^2/24) | equal? | argmax p |
|----|-------------|--------------------|--------|----------|
| 21 | 77          | 77                 | yes    | 7        |
| 22 | 84          | 84                 | yes    | 7        |
| 23 | 92          | 92                 | yes    | 8        |
| 24 | 100         | 100                | yes    | 8        |

## 5. Scope

Arithmetic is exact (Fraction), so the equality is CONCLUSIVE for every  $n \leq 5000$ . The identity and residue facts -- the parts that make Prop 7.1 hold for ALL  $n$  -- are verified symbolically-exactly, which is stronger than a numeric spot check, though a fully general proof is the paper's (this corroborates it decisively).